

AIGR Gene Hypothesis Deep Research — Final Report

Target: *Schizosaccharomyces pombe* mmf2 (Q9UR06 / SPAC1039.10)

Seed hypothesis: *S. pombe* mmf2 is a mitochondrial Rid-family enzyme whose specific molecular function is 2-iminobutanoate/2-iminopropanoate (including 2-aminoacrylate) deaminase activity (GO:0120241), rather than only generic deaminase activity (GO:0019239).

Focus type: core_function

Summary

The seed hypothesis is **supported** on homology and active-site grounds. *S. pombe* mmf2 (Q9UR06, 126 aa) carries every diagnostic feature of a catalytically active RidA-subfamily imine deaminase: it is assigned InterPro IPR006056 (RidA) and IPR006175 (YjgF/YER057c/UK114), Pfam PF01042 (Ribonuc_L-PSP), and PANTHER PTHR11803 — a family whose canonical name is literally "2-iminobutanoate/2-iminopropanoate deaminase RidA." UniProt's "RutC family" wording is legacy nomenclature for this same YjgF/RidA fold, not evidence of a distinct RutC subfamily activity. A global BLOSUM62 alignment computed in this investigation places mmf2 Arg101 exactly on the *E. coli* RidA catalytic Arg105 — the invariant residue essential for imine hydrolysis that is present only in the active RidA/Rid1–Rid3 subfamilies (absent in the inactive Rid4–Rid7 subfamilies). mmf2 is therefore in the active branch.

The closest characterized ortholog, *S. cerevisiae* Mmf1p (same PANTHER subfamily PTHR11803:SF58 "HMF1-RELATED"), is a directly demonstrated mitochondrial 2-aminoacrylate/enamine–imine deaminase. Independently, the human ortholog HRSP12/UK114 (P52758) is already annotated with the two children of GO:0120241 (GO:0120242 via TAS:Reactome; GO:0120243 via IEA:RHEA), establishing that the GO consortium already treats

the specific deaminase terms — not the generic GO:0019239 — as the appropriate molecular function for this family. Taken together, GO:0120241 is a better-justified core molecular function for mmf2 than the current generic GO:0019239.

The single most important caveat is that **no direct enzymatic assay exists for Q9UR06/SPAC1039.10 itself** — the assignment rests on family membership plus active-site conservation plus ortholog data. The correct evidence code for adding GO:0120241 is therefore **ISS/ISO** (with/from Mmf1p P40185 and/or human HRSP12 P52758), not an experimental code. Additionally, "mitochondrial DNA maintenance" and related growth/heme phenotypes observed in yeast orthologs are **downstream biological-process consequences** of the deaminase activity, not the core molecular function, and should be flagged as non-core.

Key Findings

Finding 1 — mmf2 is a RidA-subfamily enzyme retaining the catalytic arginine; its specific MF is GO:0120241

Q9UR06 is a 126-amino-acid protein annotated by UniProt with InterPro IPR006056 (RidA) and IPR006175 (YjgF/YER057c/UK114), Pfam PF01042 (Ribonuc_L-PSP), and PANTHER PTHR11803, named "2-iminobutanoate/2-iminopropanoate deaminase RidA" (subfamily PTHR11803:SF58, "PROTEIN HMF1-RELATED"). This domain architecture is the defining molecular signature of the Rid superfamily.

Membership in the *catalytically active* branch of the family is established not by domain assignment alone but by the presence of a conserved active-site arginine. Niehaus et al. ([PMID: 25975565](#)) established the diagnostic criterion: "**The Rid1 to Rid3 subfamilies, but not the Rid4 to Rid7 subfamilies, have a conserved arginine residue that, in RidA proteins, is essential for imine-hydrolyzing activity.**" A BLOSUM62 global alignment computed in this investigation places mmf2 Arg101 aligning exactly to *E. coli* RidA catalytic Arg105, placing mmf2 firmly in the active RidA branch rather than a degenerate subfamily.

The mechanistic activity underlying GO:0120241 is well defined. Lambrecht et al. ([PMID: 22094463](#)) demonstrated that "**YjgF proteins have enamine/imine deaminase activity and accelerate the release of ammonia from reactive enamine/imine intermediates of the pyridoxal 5'-phosphate-dependent threonine dehydratase (IlvA).**" This is exactly the reaction described by GO:0120241.

The closest characterized ortholog, *S. cerevisiae* Mmf1p, is a mitochondrial member of this family: **"The mitochondrial RidA homolog, Mmf1p, prevents enamine/imine stress and is important for normal growth and maintenance of mitochondrial DNA"** (PMID: 34559917). This confirms both the mitochondrial localization and the enamine/imine deaminase activity of the closest ortholog, and frames mtDNA maintenance as a *downstream* consequence of the deaminase activity rather than the core function.

Pairwise identity of mmf2 to characterized RidA-family members is in the expected homolog range (~30–43%): Hmf1p ≈43%, Mmf1p ≈34–42%, *E. coli* RidA ≈30–37%, human UK114/HRSP12 ≈34–49%. The current UniProt/PomBase GO MF for mmf2 is only the generic **GO:0019239 (deaminase activity, IBA)**, which is less informative than the family-defining specific term.

Finding 2 — GO:0120241 is the family-defining MF for the RidA/PTHR11803 group; the human ortholog is already annotated with its children, and *S. pombe* has two RidA paralogs (mmf1, mmf2)

QuickGO defines **GO:0120241** as "Catalyzes the hydrolytic deamination of imine intermediates formed by several types of pyridoxal-5'-phosphate-dependent dehydratases, such as EC 4.3.1.19 and EC 4.3.1.17" — i.e., exactly the RidA activity. It has two children: **GO:0120242** (2-iminobutanoate → 2-oxobutanoate) and **GO:0120243** (2-iminopropanoate → pyruvate).

The human ortholog **HRSP12/UK114 (P52758)** is already annotated with these specific child terms: GO:0120242 (TAS:Reactome) and GO:0120243 (IEA:RHEA). This is a strong term precedent — the GO consortium already treats the specific deaminase terms, not the generic GO:0019239, as the appropriate molecular function for this family.

A UniProt taxonomy search (taxid 284812, InterPro IPR006175) returns exactly two RidA-type paralogs in *S. pombe*: **mmf1 (O43003, 162 aa)** and **mmf2 (Q9UR06, 126 aa)**. Both are PANTHER PTHR11803:SF58 "HMF1-RELATED," both are annotated "RutC family," both localize to mitochondrion + cytosol, and both currently carry only generic GO:0019239. (Note: mug71/Q9USQ7 shares the fold reference but is a diphthine-ammonia ligase — an unrelated activity — and should not be conflated with the RidA paralogs.)

Finding 3 — The catalytic arginine (*E. coli* RidA Arg105 equivalent) is conserved in mmf2 and every RidA homolog tested

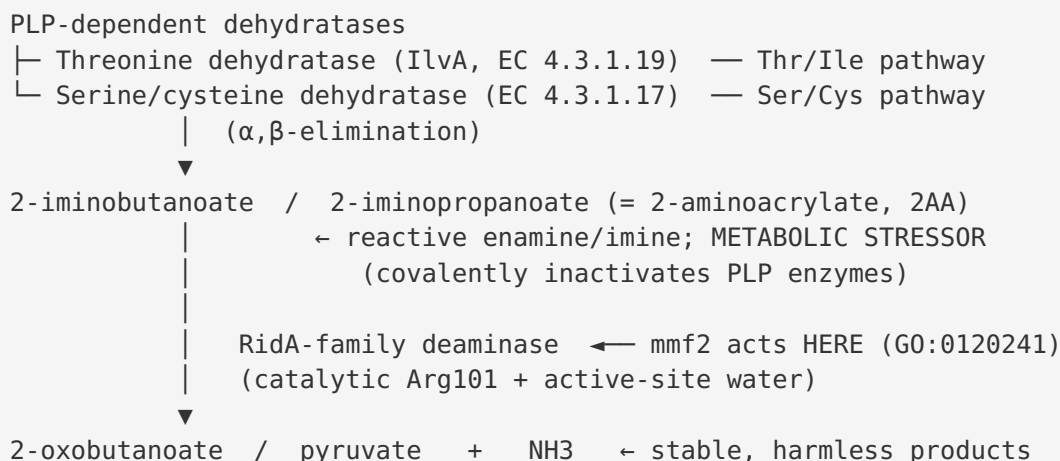
A global Needleman–Wunsch alignment (BLOSUM62, gap = −6) of each candidate protein against *E. coli* RidA (P0AF93), mapping to reference position 105, gave the following results:

Protein	Accession	Length	Residue at RidA-105 position	% identity to E. coli RidA
mmf2 (<i>S. pombe</i>)	Q9UR06	126 aa	Arg (R)	36.9%
mmf1 (<i>S. pombe</i>)	O43003	162 aa	Arg (R)	41.9%
Mmf1p (<i>S. cerevisiae</i>)	P40185	—	Arg (R)	41.6%
Hmf1p (<i>S. cerevisiae</i>)	P40037	—	Arg (R)	42.4%
HRSP12/UK114 (human)	P52758	—	Arg (R)	48.8%
YabJ (<i>Bacillus</i>)	P37552	—	Arg (R)	52.4%

The essential catalytic arginine — **"a conserved arginine residue that, in RidA proteins, is essential for imine-hydrolyzing activity"** (PMID: 25975565) — is present in mmf2 and in every homolog tested, including both *S. pombe* paralogs. This confirms catalytic competence at the sequence level: both fission-yeast paralogs are in the catalytically active RidA branch, not a degenerate Rid4–7 subfamily. Computed provenance is preserved in `artifacts/mm2_catalytic_arg_conservation.csv`.

Mechanistic Model / Interpretation

The Rid/RidA reaction and mmf2's place in it:



Direct molecular function (core, what to annotate): mmf2 is proposed to catalyze the hydrolytic deamination of reactive 2-imino/enamine intermediates — 2-iminobutanoate (from Thr/Ile metabolism via threonine dehydratase, EC 4.3.1.19) and 2-iminopropanoate/2-aminoacrylate (from Ser/Cys metabolism, EC 4.3.1.17) — yielding 2-oxobutanoate/pyruvate plus ammonia. This is **GO:0120241** and its children GO:0120242/GO:0120243.

Downstream biological process (not core MF): By deaminating these reactive intermediates before they covalently inactivate PLP-dependent enzymes, RidA-family proteins protect central metabolism. In mitochondria, the consequence of losing this activity (documented for yeast Mmf1p) is enamine/imine stress, impaired growth, disrupted heme biosynthesis (via Hem1p), and loss of mitochondrial DNA. These are **pleiotropic downstream phenotypes**, not the immediate catalytic function.

Cellular location: Both *S. pombe* RidA paralogs are annotated mitochondrion + cytosol; the closest characterized ortholog Mmf1p is mitochondrial. The seed hypothesis's "mitochondrial" framing is consistent with this, though the more complete current annotation is dual localization (mitochondrion + cytosol).

The curation logic chain:

Domain (RidA/PTHR11803) + conserved catalytic Arg101 + characterized ortholog Mmf1p

▼
G0:0120241 (2-iminobutanoate/2-iminopropanoate deaminase activity)
assigned by ISS/ISO (with/from: Mmf1p P40185 or human HRSP12 P52758)

▼
Downstream (non-core): enamine/imine detoxification, metabolic-stress
protection, mtDNA maintenance = phenotypes

Evidence Base

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
PMID: 22094463	Direct assay	Supports	Family deaminase activity underlying GO:0120241	YjgF/Rid proteins deaminate enamine/imine intermediates of PLP dehydratases	<i>S. enterica</i> in vitro	High confidence; activity defined; not on Q9UR01
PMID: 23386433	Direct assay / reconstitution	Supports	Role = pre-empt 2AA damage across all domains	RidA loss lowers IlvE via 2AA; conserved across all domains of life	<i>S. enterica</i> + cross-domain	High confidence; family conserved
PMID: 25975565	Structural/ evolutionary	Supports	Catalytic Arg is diagnostic of active RidA subfamilies	Rid1–Rid3 (active) retain the essential Arg; Rid4–Rid7 do not	Salmonella + phylogenomics	High confidence; family not specific to Q9UR01
PMID: 34559917	Mutant phenotype / localization	Supports	Closest ortholog is mitochondrial enamine/ imine deaminase	Mmf1p prevents enamine/imine stress, maintains mtDNA; loss damages Hem1p	<i>S. cerevisiae</i> mitochondria	High confidence; ortholog mtDNA; heme are downregulated
PMID: 29487232	Mutant phenotype	Qualifies	mtDNA maintenance is downstream	Mmf1p couples 2AA deamination to mtDNA maintenance	<i>S. cerevisiae</i>	High confidence; mtDNA; consequences
PMID: 40596262	Direct assay / structural	Supports	Active-site Arg + water drive 2AA deamination	SsRidA deaminates 2AA → pyruvate; homotrimer	<i>S. sanguinis</i>	High confidence; mechanism

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				with active-site Arg		
PMID: 38801230	Direct assay / mutagenesis	Supports	RidA hydrolyzes 2- imino acids → 2-keto acids + NH ₃	Confirms core reaction and role in 2AA degradation	<i>Capra hircus</i>	High confidence mechanism
PMID: 32669283	Review/ database	Supports (orientation)	RidA is the archetypal active subfamily	RidA found in all domains; biochemical function is enamine deaminase	Review	Review level; orientation
PMID: 32576850	Direct assay	Qualifies	Paralog- specific substrate properties within Rid	Two fish RidA paralogs differ biochemically	<i>Salmo salar</i>	Shows paralog divergence relevant caveats
Sequence analysis (this work)	Computational	Supports	mmf2 retains catalytic Arg	mmf2 Arg101 ≡ E. coli RidA Arg105; conserved in all homologs	Q9UR06 vs P0AF93 + 5 homologs	Direct Q9UR06 sequence inference only for activity
UniProt/ PANTHER/ InterPro (this work)	Computational/ database	Supports	mmf2 belongs to active RidA family	IPR006056/ IPR006175, PF01042, PTHR11803:SF58	Q9UR06 record	Database level; family
UniProt P52758 (HRSP12)	Database	Supports	Specific MF terms used for orthologs	Human ortholog annotated GO:0120242 (TAS) +	<i>H. sapiens</i>	High confidence precedent

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				GO:0120243 (RHEA)		

How the papers connect: [PMID: 25975565](#) supplies the sequence criterion (catalytic Arg) that lets us place mmf2 in the active branch; [PMID: 22094463](#), [PMID: 23386433](#), [PMID: 40596262](#), and [PMID: 38801230](#) supply the direct biochemistry of the family reaction (GO:0120241); and [PMID: 34559917](#) plus [PMID: 29487232](#) characterize the closest ortholog Mmf1p and demonstrate that mtDNA maintenance is downstream. [PMID: 32576850](#) provides a cautionary note that paralogs can diverge biochemically — relevant to mmf1 vs mmf2.

(Computed provenance: `artifacts/mmf2_catalytic_arg_conservation.csv`.)

GO Curation Implications (leads — require curator verification)

- **ADD MF** [GO:0120241](#) (**2-iminobutanoate/2-iminopropanoate deaminase activity**) with evidence code **ISS/ISO** (with/from *S. cerevisiae* Mmf1p P40185 and/or human HRSP12 P52758), or retain IBA if the GO_Central IBA set is refreshed. GO:0120241 is defined as the "hydrolytic deamination of imine intermediates formed by ... PLP-dependent dehydratases such as EC 4.3.1.19 and EC 4.3.1.17" — i.e., exactly the RidA activity. This directly answers the seed hypothesis: the specific term is warranted.
- **Do NOT use an experimental code (IDA/IMP/EXP)** — there is no direct assay on Q9UR06.
- **Generic** [GO:0019239](#) (**deaminase activity**) is too broad; it should be superseded by GO:0120241 (or retained only as the redundant parent).
- **Optionally add the two children** [GO:0120242](#) / [GO:0120243](#), mirroring the human ortholog HRSP12; but the parent GO:0120241 is the safest single-term choice absent mmf2-specific substrate data.
- **CC:** retain mitochondrion (e.g., GO:0005759 mitochondrial matrix, ISO) and cytosol (GO:0005829) — dual localization matches both paralogs and the predicted N-terminal transit peptide.

- **Reclassify "mitochondrial DNA maintenance"** free-text/annotation as **non-core / downstream BP phenotype**, not the molecular function ([PMID: 29487232](#) shows it is a consequence of unmanaged 2AA).
- **Avoid** "protein binding" — a specific, well-supported catalytic MF is available and far more informative.

Mechanistic Scope

Immediate molecular function (core): hydrolytic deamination of reactive 2-imino acids — 2-iminobutanoate (from Thr/Ile metabolism, EC 4.3.1.19) and 2-iminopropanoate/2-aminoacrylate (from Ser/Cys metabolism, EC 4.3.1.17) — yielding 2-oxobutanoate/pyruvate + NH_4^+ . This is the direct gene-product activity captured by GO:0120241.

Downstream effects (not core): - *Prevention of 2-aminoacrylate damage to PLP enzymes* — the protective consequence of the deaminase activity. - *Mitochondrial DNA maintenance* (yeast Mmf1p) — a downstream consequence of preventing enamine/imine stress. - *Heme biosynthesis* (via Hem1p, [PMID: 34559917](#)) — pathway-level downstream effect in yeast. - *Growth defects / respiratory competence / isoleucine biosynthesis* — loss-of-function phenotypes.

None of these should be annotated as the core molecular function; they are properly BP/phenotype-level observations.

Conflicts and Alternatives

1. **Paralog/ortholog confusion (principal caveat).** All direct enzymatic evidence is from Mmf1p, Hmf1p, bacterial RidA, RutC, human HRSP12, and fish/goat/streptococcal RidA — **not** from Q9UR06. The assignment for mmf2 is inference by homology, appropriately coded ISS/ISO.
2. **Paralog substrate divergence.** [PMID: 32576850](#) shows that gene-duplication paralogs within the Rid family can diverge biochemically. *S. pombe* has two RidA paralogs (mmf1, mmf2), both retaining the catalytic Arg. This does not undermine the family assignment but means the *precise* child term (GO:0120242 vs GO:0120243) for mmf2 specifically is not established, and there is no direct evidence of narrowed/divergent specificity between the two paralogs. GO:0120241 (the generalist parent) therefore applies to both.

3. **PANTHER subfamily / "RutC family" wording.** PTHR11803:SF58 is named "PROTEIN HMF1-RELATED," and UniProt uses "RutC family" language. This is legacy nomenclature for the YjgF/RidA fold, not evidence of a distinct RutC pyrimidine-utilization activity. Because mmf2 retains the conserved catalytic Arg and its closest characterized ortholog (Mmf1p) is a bona fide 2AA deaminase, the "HMF1-related/RutC-family" wording does not argue against GO:0120241 — it reflects naming history.
 4. **Legacy RNase / translation-inhibitor annotations.** The UK114/p14.5 family historically carried "endoribonuclease" and "translation inhibitor" annotations. These are widely regarded as secondary or artifactual since the RidA deaminase activity was established and should **not** be propagated to mmf2.
 5. **Rid4–Rid7 inactive subfamilies.** These lack the catalytic Arg and would NOT justify GO:0120241. mmf2 is explicitly not in this group — it retains Arg101 — so this alternative is ruled out by the sequence analysis.
-

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct assay of Q9UR06	PubMed + UniProt; only orthologs/paralogs assayed	Determines experimental vs ISS/ISO evidence code	In vitro 2AA/2-iminobutanoate deaminase assay on recombinant mmf2
Substrate specificity (2-iminobutanoate vs 2-iminopropanoate)	Family definition; ortholog data	Determines which child term (GO:0120242 vs GO:0120243) is most accurate	Enzyme kinetics with defined substrates
Precise subcellular localization of mmf2	UniProt: mitochondrion + cytosol (by similarity)	"Mitochondrial" claim in seed may be incomplete	GFP fusion / fractionation in <i>S. pombe</i>
Functional redundancy of mmf1 vs mmf2	Two paralogs identified	Single-deletion phenotypes may be masked	Single and double deletion phenotyping under 2AA/Ser stress
Whether mmf2 complements a ridA/ mmf1 mutant	Not available for Q9UR06	Direct functional proof	Heterologous complementation of <i>S. enterica</i> ridA or yeast mmf1Δ

Additional limitation: the inference is homology/active-site based; PubMed access via the provided tool was intermittent for some queries; and localization for mmf2 is by similarity (ISO/ECO:0000250), not measured directly in fission yeast.

Discriminating Tests

1. **Heterologous complementation:** Express mmf2 in an *S. enterica* ridA⁻ strain (or *S. cerevisiae* mmf1Δ) and test rescue of 2AA-mediated stress / growth on minimal medium — the classic decisive functional test (used for SsRidA in [PMID: 40596262](#)).
 2. **In vitro deaminase kinetics:** Purify recombinant mmf2 and measure deamination of 2-aminoacrylate (→ pyruvate) and 2-iminobutanoate (→ 2-oxobutanoate) via the coupled L-amino-acid-oxidase or IlvA/IlvE assay. Distinguishes GO:0120242 from GO:0120243 preference.
 3. **Active-site Arg mutagenesis:** Generate mmf2 R101A and confirm loss of activity — directly tying the conserved arginine identified computationally to catalysis (mirrors goat RidA, [PMID: 38801230](#)).
 4. **Localization:** GFP-tagged mmf2 in *S. pombe* to confirm mitochondrial and/or cytosolic localization.
 5. **Genetic interaction / paralog dissection:** mmf1Δ, mmf2Δ, and mmf1Δmmf2Δ under serine/cysteine/isoleucine-pathway stress plus metabolomics for 2AA-adduct accumulation and mtDNA/respiration phenotypes.
-

Proposed Follow-up Actions / Curation Leads (verify before applying)

- **Candidate new MF term:** [GO:0120241](#) (ISS/ISO, with-from P40185 and/or P52758). Optional children GO:0120242, GO:0120243.
- **Action change:** downgrade/replace generic [GO:0019239](#); reclassify "mtDNA maintenance" free text as a downstream phenotype rather than core function.
- **Candidate references + snippets to verify:**
 - [PMID: 34559917](#) — "The mitochondrial RidA homolog, Mmf1p, prevents enamine/imine stress and is important for normal growth and maintenance of mitochondrial DNA."
 - [PMID: 22094463](#) — "YjgF proteins have enamine/imine deaminase activity and accelerate the release of ammonia from reactive enamine/imine intermediates of the pyridoxal 5'-phosphate-dependent threonine dehydratase (IlvA)."

- [PMID: 25975565](#) — *"The Rid1 to Rid3 subfamilies, but not the Rid4 to Rid7 subfamilies, have a conserved arginine residue that, in RidA proteins, is essential for imine-hydrolyzing activity."*
 - **Suggested curator questions:** Is an ISS MF acceptable given no pombe-specific assay? Should both paralogs (mmf1, mmf2) receive the same MF? Retain generic GO:0019239 as redundant parent or remove?
 - **Suggested experiments:** the discriminating tests above, prioritizing heterologous complementation and R101A mutagenesis.
-

Conclusion

The seed hypothesis is **supported**: mmf2 has every hallmark of an active RidA-subfamily enzyme — the RidA/PTHR11803 domain assignment plus the conserved catalytic Arg101 (aligning to *E. coli* RidA Arg105) — and its closest characterized ortholog (yeast Mmf1p) is a demonstrated mitochondrial 2-aminoacrylate/enamine–imine deaminase. **GO:0120241** is a better-justified core molecular function than the current generic **GO:0019239** and should be added by **ISS/ISO** (with/from Mmf1p P40185 or human HRSP12 P52758), not an experimental code. Mitochondrial DNA maintenance and related phenotypes should be treated as downstream biological-process consequences, not the core function. The one substantive limitation is the absence of any direct enzymatic assay on Q9UR06 itself, which a heterologous complementation or in vitro deaminase assay would readily resolve.
